# What does a drug–drug interaction benchmark measure? Degree, embedding geometry and verified negatives on `ogbl-ddi`

Lise Lefebvre

Supervisors: Dr. Paraskevi Nousi and Dr. Maxim Samarin

DAS in Data Science, ETH Zurich

September 2026

**Abstract**

We study drug–drug interaction prediction on `ogbl-ddi`, a link prediction benchmark where drugs are nodes and known interactions are edges. We started from two questions: whether biomedical language model embeddings are better node features than the benchmark's learned ones, and whether clinically verified non-interacting pairs, taken from CRESCENDDI, are better negatives than the uniformly sampled ones the benchmark uses. Answering either required first asking what the benchmark measures. A logistic regression using only the degrees of the two drugs reaches an AUROC of 0.880, against 0.905 for a graph convolutional network with 768-dimensional PubMedBERT features: 94% of the value above chance, from two numbers per pair. The shortcut is present after a single gradient step, and it survives a change of negatives, reversing direction rather than disappearing. We also find that the embeddings, used as they come, do worse than random Gaussian vectors, because their vectors are nearly parallel and a dot-product decoder can only read angles; a whitening transformation that adds no information removes the problem and makes PubMedBERT the best input. Finally, classifiers trained without the graph show that a value on the verified negatives can be produced by recognising the drugs rather than judging the pair: a model given only the two drug identities reaches 0.979. We report the controls that separate these effects, and the one result none of them explains.

## 1 Introduction

Two drugs interact when taking them together produces an effect that is different from what each of them does alone. Some interactions are useful, many are

harmful, and the risk grows with the number of drugs a patient takes at once. Finding them experimentally is not possible at scale: with the 4,267 drugs of the dataset we use, there are more than nine million pairs, and each one would have to be studied separately. This is why interactions are predicted computationally, and why a prediction is most useful for the pairs nobody has looked at yet.

A natural way to state the problem is as a graph. Each drug is a node, each known interaction is an edge, and predicting an interaction means predicting a missing edge. `ogbl-ddi`, one of the link prediction datasets of the Open Graph Benchmark [6], is built exactly this way, and it is the dataset we use. The standard approach learns one vector per drug during training and scores a pair by combining the two vectors.

This project began with two questions about that approach.

The first concerns the node features. The benchmark gives the drugs no features at all, so each method invents its own. Since a drug has a text description and a chemical structure, an obvious idea is to use a pretrained language model: PubMedBERT [4] for the description, ChemBERTa [1] for the structure. Do these features help?

The second concerns the negatives. To train and to evaluate, a model needs pairs that do not interact. The benchmark takes pairs that are not edges, sampled uniformly, and treats them as negatives. But a pair can be absent from the graph because the two drugs do not interact, or because nobody has ever studied them together. CRESCENDDI [9] is a resource where clinical sources have looked at the pairs, so its negatives are checked rather than assumed. Does using them change the picture?

Answering either question turned out to require a third one first: what does the standard evaluation measure? Our results say that on this dataset it measures, to a large extent, how many interactions each of the two drugs already has. A logistic regression with two features, the degrees of the two drugs, reaches an AUROC of 0.880 where our GCN reaches 0.905. All values we report are AUROC, the probability that a positive pair is scored above a negative one. The reason is in the data: the uniformly sampled negatives involve drugs that are much less studied than the ones in real interactions, so "both drugs are well connected" is already a good rule.

Once this is known, the two original questions have answers, and they are more specific than a yes or a no. The embeddings, used as they come, are worse than random numbers, for a reason that has nothing to do with what they contain: their vectors all point in nearly the same direction, and a dot-product decoder can only see directions. A fixed linear transformation that adds no information removes the problem and makes them the best input we have. The verified negatives do not remove the degree effect either; they reverse it, because the drugs they involve are more connected than average. What they do give is stability: a model trained on them behaves almost the same on every evaluation set we tried.

We also found that a high value on the verified negatives can be obtained without knowing anything about the drugs. A model given only the identities

of the two drugs reaches an AUROC of 0.979 on them, because the same drugs appear in the training and the test pairs. This is not a property of the model but of how the evaluation is built, and it is easy to check for.

The contribution of this report is therefore less a method than a set of measurements and the controls that make them readable: a degree-only baseline, a degree-matched negative set, random vectors as a feature baseline, a model that sees only drug identities, and a split in which no test drug is seen during training. Each removes one way of scoring well without answering the question. One result survives all of them, and we say which.

The report is organised as follows. Section 2 describes the data: the `ogbl-ddi` graph and its official protocol, then CRESCENDDI, how its pairs map onto the graph, and the degree structure of the resulting evaluation sets, which is where the main effect of this report begins. Section 3 gives the experimental setup, the two classifiers we use as a control, the drug-disjoint split, the leakage risks and what we did about each, and the definition of the three evaluation sets. Section 4 reports the results, in six steps: a reproduction of the benchmark, a model using only the degree, the GCN on the three evaluation sets, the effect of training on verified negatives, a comparison of node features that leads to the whitening experiment, and the classifiers trained without the graph. Section 5 draws the results together around the distinction between signals that describe a single drug and signals that describe a pair. Section 6 states what our results do not show, and Section 7 what we would do next.

# 2 Data

## 2.1 The OGB-ddi dataset

`ogbl-ddi` is one of the link prediction datasets of the Open Graph Benchmark [6]. It is an undirected and unweighted graph. Each node is a drug, approved or experimental, and each edge means that taking the two drugs together has an effect that is clearly different from what we would expect if they acted separately. The interactions come from DrugBank [13].

The graph has 4,267 nodes and 1,334,889 edges. That is a density of 14.7%: almost one drug pair out of seven is recorded as interacting. This is much denser than the other link prediction datasets of the benchmark, and it will matter for our results. The degrees are also very unequal. Computed on the training edges, the least connected drug has one interaction, the median drug has 446, and the most connected has 2,234, which is more than half of all the drugs in the graph.

The dataset comes with no node features. Every method has to choose its own way of representing the drugs. The standard baselines of the benchmark learn one embedding vector per node during training. We do something else and use PubMedBERT embeddings of the drug descriptions. This is our choice and not part of the benchmark.

**Split.** The edges are divided 80/10/10 into training, validation and test. The split is not random. OGB uses a *protein-target split*: the edges are separated according to the proteins the drugs bind to in the body. The drugs in the test set therefore mostly target different proteins from those in the training set. The idea is to stop a model from succeeding just because a very similar drug was already in the training data.

**Evaluation.** The official metric is Hits@20. Each true test edge is ranked against about 100,000 randomly sampled non-interacting pairs, and the metric is the proportion of true edges that end up in the top 20. This is demanding: a pair has to beat almost all of the 100,000 negatives, not just a typical one.

We use AUROC instead. Hits@20 depends on how many negatives you rank against, and our three evaluation sets have very different sizes: 133,117 random negatives against 811 CRESCENDDI pairs. Being in the top 20 out of 811 and in the top 20 out of 133,117 are not the same thing, so the values would not be comparable. AUROC is the probability that a positive edge gets a higher score than a negative one, and it does not depend on the sizes. It is also what most of the works we compare with report on this dataset.

The negatives used in the official evaluation are sampled uniformly among the pairs that are not edges, and nobody checks them. A pair counts as negative because it is absent from the graph. That can mean the two drugs do not interact, or simply that nobody has ever studied the question. What follows from this choice is the subject of Section 4. Table 1 collects the counts of the frozen dataset used throughout this report: the graph before and after the removal of the conflicting edges described below, the three splits of positive edges, the CRESCENDDI pairs and their portions, and the three sets of negatives used for evaluation.

## 2.2 CRESCENDDI and its relation to OGB-ddi

CRESCENDDI is a set of drug pairs that clinical sources have looked at [9]. It was built to test signal detection methods in pharmacovigilance, not for machine learning, so we have to be careful about what it says.

Its rows are not pairs but triplets: two drugs and one adverse event. There are 10,286 positive controls and 4,544 negative controls, over 454 drugs and 179 events. A triplet is a positive control if the pair appears in all three sources the resource is built from, which are the British National Formulary, the French ANSM Thesaurus and Micromedex, and if the text describes a real clinical effect. Descriptions that only propose a mechanism were left out. A triplet is a negative control if the pair appears in none of the three sources, and if a PubMed query for the two drugs together with the event returns nothing.

So a negative control does not mean that the two drugs do not interact. It means that nobody has published anything about them causing that particular event. The authors say this themselves. It is still much better than what the benchmark does, where a pair is a negative simply because it is not in the graph.

| | | Count | Mean degree |
|---|---|---|---|
| *Graph* | | | |
| Drugs (nodes) | | 4,267 | |
| Edges before conflict removal | | 1,334,889 | |
| Edges removed (CRESCENDDI conflicts) | 2,233 | (0.167%) | |
| Edges after removal | | 1,332,656 | |
| Graph density | | 14.6% | |
| *Positive edges (after removal)* | | | |
| Training | | 1,066,276 | |
| Validation | | 133,263 | |
| Test | | 133,117 | 793.8 |
| *CRESCENDDI* | | | |
| Pairs mapped to OGB-ddi | | 4,049 | 966.9 |
| of which conflicting with OGB | 2,233 | (55.1%) | |
| Training portion | | 2,834 | |
| Validation portion | | 404 | |
| Test portion | | 811 | 966.9 |
| *Negative sets used for evaluation* | | | |
| Random non-edges | | 133,117 | 453.8 |
| Degree-matched non-edges | | 810 | 962.2 |
| CRESCENDDI verified | | 811 | 966.9 |
| *Node features* | | | |
| PubMedBERT dimension / coverage | 768 | / 100% | |

Table 1: Main counts of the frozen dataset `frozen_r5`. All degrees are computed on the training edges only, so that no test information enters any reported number. One CRESCENDDI test pair could not be matched by the degree-matched construction, which therefore has 810 pairs instead of 811.

We only need pairs, so we drop the event column. Pairs that appear several times, once per event, become duplicates and we remove them. We then map the drug names to DrugBank identifiers and the identifiers to OGB-ddi nodes. This leaves 4,743 positive pairs out of 10,286 triplets, and 4,049 negative pairs out of 4,544. Five drug names could not be mapped. The two sets do not share a single pair, which is what the construction rules predict.

**How many of these pairs are in OGB-ddi?**

We then looked at how many of these pairs are edges in OGB-ddi. We cannot compare the result to the density of the graph, which is 14.7%. The drugs in CRESCENDDI are used a lot in the clinic and are therefore very connected, and two connected drugs are more likely to be an edge whatever a compendium says about them. So for each set we also built a null: random pairs drawn from

the same degree bins, with no clinical information at all. This is the same idea as the degree-matched control of Section 3.5.

Table 2 reports the share of CRESCENDDI pairs that are edges in OGB-ddi, next to that null.

| Set | $n$ | In OGB-ddi | Null | Difference |
|---|---|---|---|---|
| CRESCENDDI positives | 4,743 | 97.4% | 70.2% | +27.2 |
| CRESCENDDI negatives | 4,049 | 55.1% | 59.8% | −4.7 |

Table 2: Share of CRESCENDDI pairs that are edges in OGB-ddi, next to a null of random pairs with the same degrees. Measured on the original graph, before the edges described below were removed.

The first thing this table tells us is that our mapping works. If we had made a mistake going from drug names to node indices, the positive pairs would not come back as edges 97.4% of the time.

The second thing is more interesting. The two rows do not behave the same way. The positive pairs are edges 27 points more often than their degree alone would give, so OGB-ddi finds almost all the interactions that clinicians have confirmed. The negative pairs are edges 4.7 points less often than their degree would give. The sign is the one we want, but the effect is small. If we treat "being an edge in OGB-ddi" as a test for "is this a real clinical interaction", the test almost never misses a true one, and almost never rejects a false one.

This comes from how the graph was made. OGB-ddi comes from DrugBank [13], which contains interactions predicted from the mechanism of the drugs as well as interactions seen in patients. Being plausible is a much easier condition than being observed. The graph is therefore nearly complete on real interactions, and says very little about pairs that do not interact. The null says the same thing from the other side: between drugs this connected, most pairs are edges anyway.

**Removing the conflicting edges**

2,233 of the 4,049 negative pairs, that is 55.1%, are edges in OGB-ddi. A pair cannot be a positive edge in the graph and a negative in our evaluation set at the same time, so we have to choose. We remove these 2,233 edges from the graph and keep all 4,049 pairs as negatives. This removes 0.167% of the edges. We do not re-split afterwards, so the edges are taken out of whichever official split they were in.

The drugs involved in these conflicts are not unusual ones. The twenty most frequent are ordinary, widely prescribed medicines (acetaminophen, lorazepam, levothyroxine, lisinopril) with degrees between 1,000 and 2,200, against a mean of about 625 in the graph. Across the 430 drugs appearing in the CRESCENDDI negatives, the share of a drug's pairs that conflict is strongly correlated with its degree (Spearman $\rho = 0.68$). A conflict is therefore mostly a consequence of

how much a drug has been studied, rather than of a disagreement about that particular drug.

The 2,233 removed edges fall in the official split as 1,635 training, 226 validation and 372 test. Edges are removed in place rather than triggering a re-split, so each remaining edge keeps its original assignment. The three splits are correspondingly smaller than the official ones: our test set contains 133,117 positive edges instead of 133,489.

We are not removing these edges because we have shown they are wrong. Table 2 shows they appear at roughly the rate the degree alone predicts. We remove them because our evaluation needs the labels to be consistent, and because between the two sources the clinical one is the one we want to be measured against.

| Set | $n$ | Mean degree | Median degree |
|---|---|---|---|
| Random negatives | 133,117 | 453.8 | 368 |
| Positive edges | 133,117 | 793.8 | 788 |
| Degree-matched | 810 | 962.2 | 925 |
| CRESCENDDI | 811 | 966.9 | 925 |

Table 3: Mean and median degree of the drugs in each evaluation set.

Two things come out of Table 3.

First, the degree-matched control works. Its mean degree (962.2) is almost the same as the CRESCENDDI one (966.9), and the medians are identical (925). This is what lets us separate the effect of the degree from everything else. The two sets are not identical in every other respect: the CRESCENDDI pairs have been verified in clinical compendia, and they are also built from a particular set of drugs, which we come back to in Section 4.6.

Second, and this is the important part, the sets are not in the order one would expect. The random negatives have a low degree (453.8) because most drugs in the graph have few interactions, so a uniform sample mostly picks rarely used drugs. The positive edges are higher (793.8) because real interactions involve drugs that have been well studied. But the CRESCENDDI pairs are higher still (966.9), because they are made of drugs that are used often in the clinic and have therefore been tested against many others.

This has a direct consequence. Take a model that only looks at the degree and predicts an interaction when both drugs are well connected. On random negatives it looks good, because the positives (793.8) are more connected than the negatives (453.8). On CRESCENDDI the same model looks bad, because now the negatives (966.9) are more connected than the positives (793.8), so its ranking is reversed. On CRESCENDDI the degree is not a weak signal, it is a signal that points the wrong way.

The same rule therefore gives a good value on one evaluation set and a value below chance on the other, without the model changing at all. We test this directly in Section 4.

# 3 Methods

## 3.1 Experimental setup

The encoder is a 3-layer GCN [8] with hidden dimension 256 and dropout 0.5. The score of a pair is the dot product of the two node representations. The decoder has no parameters of its own, so everything the model learns is in the representations. We train with `BCEWithLogitsLoss` and Adam (lr $= 5 \times 10^{-4}$) for 400 epochs, in full batch, with gradient clipping. Node features are PubMedBERT embeddings of the drug descriptions [4], 768 dimensions, covering all 4,267 drugs.

All results are averaged over 5 seeds. The seed controls the initialisation of the model, the draw of the training negatives, and, where it applies, the subsample of positive edges. The evaluation sets are the same for every seed, so the standard deviations we report describe variability in training, not variability in what is being measured.

We report the value at the last epoch, not at the epoch chosen on validation. Our validation set uses random negatives, and performance there stops improving after about 15 epochs, while performance on the two control sets keeps improving until around epoch 300. Choosing the epoch on validation would therefore stop training long before the control sets have converged. Using a fixed budget also means that no configuration is favoured by a lucky stopping point. Epoch 400 is itself an arbitrary choice; what matters is that it is the same for every run.

## 3.2 Classifiers on the node features

The degree-only model of Section 4 uses the structure of the graph and nothing about the drugs. We also want the opposite: models that see the drug representations and no structure at all. We therefore train two classical classifiers directly on the node features, with no message passing.

A pair has to be represented by a single vector, and that vector has to be symmetric, since the graph is undirected and $(u, v)$ and $(v, u)$ are the same pair. We use the elementwise product $x_u \odot x_v$. A logistic regression on this representation computes $\sum_k w_k \, x_{uk} x_{vk}$, which is a dot product with one learned weight per coordinate, so it is the closest no-graph counterpart of the decoder used by the GCN. We also train a random forest (200 trees, maximum depth 25, at least 10 samples per leaf), which can combine coordinates in ways a linear model cannot.

The two training regimes are the ones of Section 4: either random negatives in a 1:1 ratio, or the CRESCENDDI training pairs with the positives subsampled to 10:1 and the loss reweighted so that both classes contribute equally. Training uses 100,000 positive pairs in the random regime, rather than all 1,066,276, to keep the feature matrices within memory. Evaluation is identical to the one used for the GCN: the same three negative sets, and the positives subsampled to the size of each negative set.

Two models are included as references rather than as candidates. The degree-only logistic regression of Section 4, run through this same pipeline, recovers the values of Table 4 and shows what the structure alone gives. Random Gaussian vectors, used as node features, show what a model of the same family obtains with no information about the drugs at all. This is the floor against which the embeddings have to be read, and on the official split it lies well above 0.5, which is the reason for the split described next.

### 3.3 A drug-disjoint split

Section 3.4 notes that 807 of the 811 CRESCENDDI test pairs are made of drugs that also appear in the training pairs, so a model can score them by recognising the drugs rather than by judging the pair. To measure how much this allows, we build a split in which the two cannot be confused.

Every drug is assigned to one of two groups of equal size, balanced on degree, with the CRESCENDDI drugs and the remaining drugs split separately. A pair is used for training only if both its drugs are in the first group, and for testing only if both are in the second; pairs with one drug in each group are discarded. This applies to the positive edges and to the negatives of both regimes, so no drug of a test pair appears in any training pair. All 4,049 CRESCENDDI pairs are pooled and split again in this way, and the degree-matched set is rebuilt for the resulting test pairs, drawing only from the second group. No validation portion is needed, since we report a fixed epoch and select nothing. About 49% of the CRESCENDDI pairs survive the filter, leaving between 1,004 and 1,101 test pairs and between 927 and 978 training negatives, depending on the split. We report means and standard deviations over three splits, so the variability covers which drugs fall on which side.

The degrees used as features, and for the degree matching, are still computed on all training edges of the graph. The degree is a property of the graph, not of the pairs given to the classifier, and keeping it unchanged is deliberate: it lets us show that a drug-disjoint split removes the recognition of drugs without removing the degree effect.

As a check that the split is watertight, we train a logistic regression on the drug identities alone, that is on the sum of the two one-hot vectors of the pair. Such a model has no information about a drug it has not seen, so it must give the same score to every test pair. It returns an AUROC of exactly 0.500 on all three evaluation sets, in both regimes.

### 3.4 Data leakage

Leakage happens when information from the test set reaches the model during training. The model then looks better than it is. Link prediction is particularly exposed, because the edges are both the input and the labels [14]. Below are the risks in our setup and what we did about each.

**1. The encoder seeing the edge it has to predict.** In a GCN an edge can be used in two ways: as part of the graph for message passing, or as a label for the decoder. It must not be both. If the encoder aggregates over edge $(u, v)$, then the representations of $u$ and $v$ already contain the fact that they are connected, and asking the decoder to score that pair is not really a prediction. We use only the training edges for message passing. The validation and test edges are scored by the decoder but never added to the graph.

**2. Random negatives falling on CRESCENDDI pairs.** We sample random negatives uniformly among node pairs, and nothing stops a draw from landing on a pair that is also in CRESCENDDI. If that happens, part of our evaluation set ends up in the training data. We therefore exclude all graph edges and all 4,049 CRESCENDDI pairs when sampling, and we do the same for the degree-matched control. In practice this removes almost nothing, since there are about 9 million possible pairs, but it makes the three evaluation sets exactly independent rather than approximately independent.

**3. Using CRESCENDDI both to choose the model and to report the result.** At first we used the CRESCENDDI pairs to pick the training epoch and also to report the final value. That is not correct: if we stop at the epoch that gives the best CRESCENDDI result, the CRESCENDDI result we report is too optimistic. We now split the pairs into three disjoint parts (70% train, 10% validation, 20% test). The test part is scored once, at the end. The validation part is only used to follow training; nothing is selected on it, since we report a fixed epoch for every run (Section 3.1).

**4. The same drug appearing in train and test.** The 4,049 CRESCENDDI pairs are built from only 430 distinct drugs, so the same drug appears in many pairs. With a split at the level of pairs, 99.2% of the drugs in the test portion also appear in the training portion, each one 15.4 times on average, and 807 of the 811 test pairs are made of two drugs that were both seen during training. A shortcut of the form "this drug is usually safe" is therefore fully available to the model.

How much this allows can be measured, and it is more than we first assumed. Trained on the CRESCENDDI pairs, a random forest whose features are random vectors, containing nothing at all about the drugs, reaches an AUROC of $0.822 \pm 0.005$ on the CRESCENDDI test pairs, and a logistic regression given only the identities of the two drugs reaches 0.979. Both fall to about 0.500 once the drugs of the test pairs are held out of training (Section 4.6). The same models stay near or below chance on the degree-matched set, whose drugs are not the ones seen during training.

A high value on the CRESCENDDI set alone can therefore be obtained with no knowledge of the drugs at all, and we rely on the degree-matched set for our claims about what a model has learned. Section 3.3 describes the drug-disjoint split we use for the classifiers; the same split for the GCN is left to future work.

**5. Two risks we cannot remove.** The first comes from the pretrained language model. PubMedBERT was trained on biomedical abstracts [4], and the DrugBank interactions were curated from that same literature. Some of the edges we ask the model to predict may therefore already be in its input features. This is even more direct for the drug descriptions, which sometimes name the interacting drug explicitly. Embeddings computed from the chemical structure, such as ChemBERTa [1], would not have this problem, since molecular structure is not where interaction reports are written down. We compare the two in Section 4.5. The comparison does not settle the leakage question, because both embeddings turn out to be worse than random vectors for reasons unrelated to their content, so the risk described here remains unmeasured.

The second is the transductive setting. All the drugs are already in the graph when we train, so the model never has to predict for a drug it has not seen [5]. This is the standard protocol for `ogbl-ddi` and it lets us compare with published results, but it limits what our numbers say about new drugs. We come back to this in Section 6.

### 3.5 The three evaluation sets

Every model in this report is evaluated on the same 133,117 positive test edges, against three different sets of negatives.

- **Random**: 133,117 non-edges sampled uniformly. This is the standard protocol.
- **Degree-matched**: for each CRESCENDDI test pair, a non-edge whose two drugs have the same degrees. Nobody has verified these pairs; they are ordinary non-edges taken from the same region of connectivity.
- **CRESCENDDI**: the 811 verified pairs of the test portion.

Going from the first to the second changes the degree. Going from the second to the third changes the clinical verification, and also which drugs the pairs are made of, since the CRESCENDDI pairs come from the 430 drugs of the compendia; we return to this in Section 4.6.

To build the matched set, we bin the nodes by degree on a logarithmic scale, and for each CRESCENDDI test pair we draw a pair of nodes from the same two bins, rejecting any candidate that is an edge, a CRESCENDDI pair, or already drawn. One pair out of 811 could not be matched.

## 4 Results

### 4.1 Reproduction of the benchmark

Under the standard protocol, with random negatives for training and for evaluation, our GCN reaches an AUROC of $0.905 \pm 0.003$ over 5 seeds. For reference, Li et al. [11] report an AUC of 0.900 on the same dataset. The two values are

not directly comparable: LLM-DDI uses five-fold cross-validation instead of the official protein-target split, an MLP decoder instead of a dot product, and a preprocessed graph whose statistics differ from the released one. We give their value only to show roughly where ours sits.

What matters for the rest of this section is not that our model is the best one. It is that it is an ordinary one, trained in the usual way and evaluated in the usual way, with a value in the usual range. So what follows cannot be blamed on a weak model.

## 4.2 A model using only the degree

We now ask how much of that value can be obtained without knowing anything about the drugs. We train a logistic regression with two features: the degrees of the two drugs of the pair. Since the graph is undirected we use the minimum and the maximum of the two, so that the prediction does not depend on the order in which a pair is written, and we take the logarithm of both because the degree distribution is very skewed. The degrees are computed on the training edges only. The model is trained on the same 1,066,276 positive edges and the same number of random negatives as the GCN.

| Test negatives | AUROC |
|---|---|
| Random | 0.8799 |
| Degree-matched | 0.3280 |
| CRESCENDDI | 0.3289 |

Table 4: Degree-only logistic regression: two features per pair, no chemical and no textual information.

Three things follow from Table 4.

First, this model reaches 0.8799 under the standard protocol, against 0.905 for the GCN with 768-dimensional embeddings and three convolution layers. That is 97.2% of the GCN's value, or 94% of it measured above chance, since an AUROC of 0.5 is what a coin flip gives. Almost everything the benchmark measures is available from two numbers per pair.

Second, the same model drops to 0.328 and 0.329 on the two other sets. A value below 0.5 does not mean the model has no information; it means its ranking is reversed. The rule it learned, that two well connected drugs are likely to interact, puts the verified negatives above the true positives, because those negatives have a higher mean degree (966.9) than the positives (793.8). The coefficients confirm this: both are positive, +2.42 on the smaller degree of the pair and +0.67 on the larger one. The bigger weight on the smaller degree makes sense given how the graph was built. A pair is only recorded as interacting when both drugs have been studied enough for someone to notice, so it is the less studied of the two that limits the prediction.

Third, the values on the degree-matched set (0.3280) and on CRESCENDDI (0.3289) differ by 0.0009, and the mean scores given to the negatives of the two

sets are 0.7929 and 0.7932. This is what should happen: the two sets have the same degrees by construction and the model sees nothing else, so it cannot tell them apart. We take this as a check that the control is correctly built. It also means that any difference between these two sets for another model cannot come from the degree.

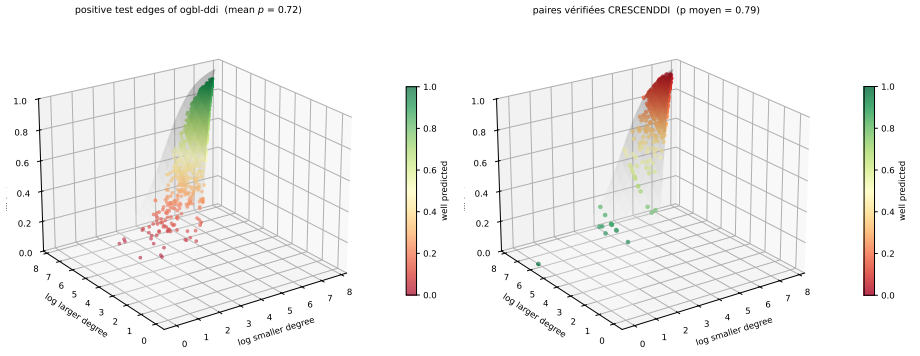



Figure 1: The degree-only model drawn as a surface over the two degrees of a pair, with two evaluation sets on it. Left: the positive test edges, where a high probability is the correct answer. Right: the CRESCENDDI verified negatives, where a low probability is correct. Colour marks correctness in both panels, so green is right and red is wrong, but it is inverted on the right because those pairs are negatives. Points are subsampled.

Figure 1 shows what this model is. It has only two numbers per pair, so its prediction is a single surface over the degree plane, and every point sits exactly on it.

The surface is the same in both panels. Only the region occupied by the points changes. The positive edges are high on the surface, which is the right answer for them. The CRESCENDDI pairs are higher still, because their drugs are more connected, but they are negatives, so the same answer is now wrong.

It is one rule, applied to two sets that sit in different parts of its input. That is enough to make it right on one and wrong on the other.

## 4.3 The GCN on the three evaluation sets

Table 5 puts the two models side by side on the three evaluation sets.

The GCN beats the degree baseline on all three sets, but by very different amounts. On the standard evaluation it gains 0.02. On the two control sets it gains more than 0.15. Whatever the GCN uses beyond the degree is almost invisible in the benchmark number.

Where does the shortcut come from? After a single gradient step, the GCN already gives 0.863, 0.301 and 0.310 on the three sets, which is essentially the profile of the degree model. This comes from message passing itself: the repre-

| Model | Random | Degree-matched | CRESCENDDI |
|---|---|---|---|
| Logistic regression (2 features) | 0.8799 | 0.3280 | 0.3289 |
| GCN + PubMedBERT (768 dim) | $0.905 \pm 0.003$ | $0.637 \pm 0.009$ | $0.485 \pm 0.019$ |

Table 5: AUROC on the three evaluation sets. The GCN values are means and standard deviations over 5 seeds at epoch 400. The logistic regression is deterministic given its training set.

sentation of a node is an aggregation over its neighbours, so its norm grows with the number of neighbours, and the dot product of two vectors grows with their norms. Well connected drugs therefore get high scores whatever the weights have learned. The shortcut is there before any learning.

The two quantities also converge at very different speeds. The value on random negatives reaches 0.88 after 25 epochs and barely moves afterwards. The value on the degree-matched set is still 0.36 at epoch 25, 0.58 at epoch 200, and only settles around epoch 300. A training budget chosen on benchmark performance stops long before the model has learned what the control sets measure.

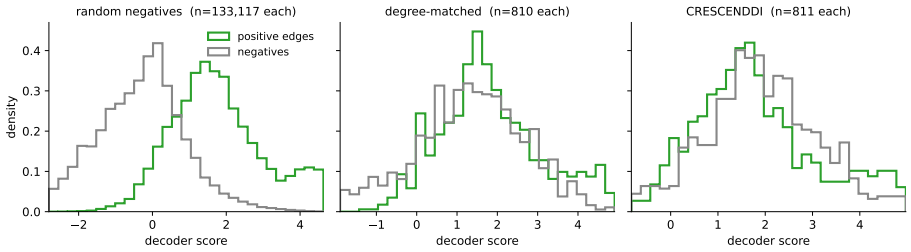



Figure 2: Distribution of the decoder scores for the model trained on random negatives, one panel per evaluation set (seed 0, epoch 400). Raw dot products, before the sigmoid. The positives are subsampled to the size of each negative set, so the two classes have equal counts in every panel. Axes are restricted to the bulk of each distribution.

Figure 2 shows what these AUROC values look like. On the standard evaluation the two classes form two separate modes, and the model separates them easily. On the degree-matched set they already overlap heavily. On CRESCENDDI the two distributions lie almost exactly on top of each other.

This is worth stating precisely, because an AUROC of 0.50 can mean two different things. It can mean that the model ranks the classes in the wrong order, or that it does not distinguish them at all. Here it is the second: the model is not giving the verified negatives low scores, it is giving them the same scores as the true interactions. There is no threshold that would separate them.

### 4.4 Training on verified negatives

Everything above used random negatives during training. We now train the same model on the CRESCENDDI training pairs instead. There are only 2,834 of them against more than one million positive edges, a ratio of about 380 to 1. A loss weight alone cannot handle that, since it cannot create signal out of 2,834 examples, so we first subsample the positives to 10:1 and then reweight the loss so that the two classes contribute equally. The encoder still uses the full graph; only the pairs given to the decoder change.

Table 6 reports the GCN and the degree-only model in both training regimes, on the same three evaluation sets.

| Training negatives | Model | Random | Degree-matched | CRESCENDDI |
|---|---|---|---|---|
| Random | GCN + PubMedBERT | $\mathbf{0.905 \pm 0.003}$ | $0.637 \pm 0.009$ | $0.485 \pm 0.019$ |
| | Degree only (2 feat.) | 0.880 | 0.328 | 0.329 |
| CRESCENDDI | GCN + PubMedBERT | $0.659 \pm 0.005$ | $\mathbf{0.739 \pm 0.027}$ | $0.687 \pm 0.016$ |
| | Degree only (2 feat.) | $0.147 \pm 0.001$ | $0.695 \pm 0.010$ | $0.684 \pm 0.003$ |

Table 6: AUROC depending on the negatives used for training and for evaluation, for the GCN and for the degree-only logistic regression of Section 4. GCN values are means and standard deviations over 5 seeds at epoch 400. The degree-only model is deterministic in the random regime; in the CRESCENDDI regime it depends on the subsample of the positives, and is averaged over 5 seeds.

Training on CRESCENDDI does not remove the degree effect, it reverses it. The verified negatives are more connected than the positive edges, so a model trained on them learns that two well connected drugs are unlikely to interact. The degree-only model trained this way shows what that rule gives on its own: 0.684 on CRESCENDDI and 0.695 on the degree-matched set, against 0.147 on the random negatives, where it points the wrong way. Measured against this baseline, the GCN does not do better on CRESCENDDI (0.687 against 0.684) and only slightly better on the degree-matched set (0.739 against 0.695). Where it clearly differs is on the random negatives, 0.659 against 0.147: it has not simply learned the reversed rule. It is also far more stable. Its three values are within 0.080 of each other, against 0.548 for the degree-only model trained on the same pairs and 0.420 for the GCN trained on random negatives.

One detail is worth pointing out. The model trained on CRESCENDDI does better on the degree-matched set (0.739) than on the CRESCENDDI set it was actually trained on (0.687). The difference is 0.052 against a standard deviation of 0.027, so it is suggestive rather than established.

The training curves show the shortcut being reversed. In the run trained on CRESCENDDI, the value on random negatives falls from 0.863 at epoch 1 to 0.548 at epoch 50, before recovering to 0.659 by the end, while the value on CRESCENDDI rises over the same interval. From about epoch 100 the three

curves go up together (Figure 3).

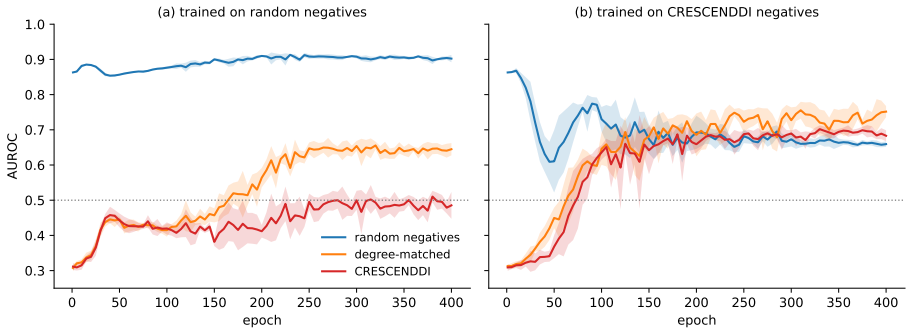



Figure 3: AUROC during training, mean over 5 seeds with a ±1 standard deviation band. The dotted line marks chance. Left: trained on random negatives. Right: trained on CRESCENDDI negatives.

## 4.5 Do the node features help?

Everything so far used PubMedBERT embeddings as node features. We now ask a simpler question: do these features help at all? We compare several inputs to the same encoder, keeping everything else fixed.

- **Random vectors**: 768 numbers drawn from a standard normal for each drug, the same draw for every run. They contain nothing about the drugs.
- **ChemBERTa** [1]: 768 dimensions computed from the chemical structure of the molecule.
- **PubMedBERT** [4]: 768 dimensions computed from the DrugBank text description. This is the configuration used elsewhere in this report.
- **GPT**: the 1,536-dimensional `text-embedding-ada-002` vectors of the drug descriptions, taken from the DDI-LLM framework [11]. These cover 81.6% of our drugs; the remaining 786 nodes receive the mean vector.
- **The same embeddings after whitening**, described below.

Each is run in both training regimes, on 5 seeds, for 400 epochs, and Table 7 reports the result.

### The embeddings as they come

Take the first four rows of each block. On the standard evaluation they are almost the same, between 0.905 and 0.927. The benchmark cannot tell apart a model that was given the chemistry of the drugs from one that was given 768 random numbers. This is the same conclusion as Section 4, reached a third way.

| Features | Random | Degree-matched | CRESCENDDI |
|---|---|---|---|
| *Trained on random negatives* | | | |
| Random vectors | $0.927 \pm 0.004$ | $0.831 \pm 0.003$ | $0.625 \pm 0.009$ |
| ChemBERTa | $0.908 \pm 0.005$ | $0.690 \pm 0.014$ | $0.536 \pm 0.018$ |
| GPT (ada-002) | $0.905 \pm 0.003$ | $0.656 \pm 0.021$ | $0.504 \pm 0.012$ |
| PubMedBERT | $0.905 \pm 0.003$ | $0.642 \pm 0.015$ | $0.490 \pm 0.023$ |
| ChemBERTa whitened | $0.934 \pm 0.003$ | $0.860 \pm 0.011$ | $0.639 \pm 0.008$ |
| PubMedBERT whitened | $\mathbf{0.941 \pm 0.004}$ | $\mathbf{0.871 \pm 0.014}$ | $\mathbf{0.645 \pm 0.014}$ |
| *Trained on CRESCENDDI negatives* | | | |
| Random vectors | $0.739 \pm 0.009$ | $0.873 \pm 0.004$ | $0.795 \pm 0.007$ |
| ChemBERTa | $0.665 \pm 0.007$ | $0.777 \pm 0.015$ | $0.690 \pm 0.012$ |
| GPT (ada-002) | $0.666 \pm 0.005$ | $0.764 \pm 0.012$ | $0.680 \pm 0.007$ |
| PubMedBERT | $0.660 \pm 0.004$ | $0.744 \pm 0.028$ | $0.687 \pm 0.017$ |
| ChemBERTa whitened | $0.756 \pm 0.006$ | $0.882 \pm 0.006$ | $0.796 \pm 0.006$ |
| PubMedBERT whitened | $\mathbf{0.757 \pm 0.006}$ | $\mathbf{0.889 \pm 0.006}$ | $\mathbf{0.795 \pm 0.008}$ |

Table 7: AUROC for six feature types, means and standard deviations over 5 seeds at epoch 400. The GPT row uses 1,536 dimensions and the others 768, which matters for the comparison; see the discussion of capacity below. The PubMedBERT rows come from a separate batch of runs from Table 6; the small differences between the two tables are run-to-run variation.

On the two control sets they are not the same at all, and the order is the opposite of what one would expect. The random vectors are the best of the four everywhere, by 0.19 on the degree-matched set in the first regime and by 0.10 in the second. These differences are many times the standard deviation across seeds.

**Why the embeddings do worse**

Our first idea was that the embeddings might be badly scaled. The loss at the first epoch is about 0.4 for the random vectors, 4 for PubMedBERT and 15 for ChemBERTa, which suggests the model starts far outside a usable range. But the vector lengths do not support this: the mean norm is 27.7 for the random vectors, 26.7 for ChemBERTa and 15.1 for PubMedBERT, so the embeddings are not longer, and PubMedBERT is the shortest of the three.

The explanation is in the angles. Our decoder is a dot product,

$$s_{uv} \;=\; z_u \cdot z_v \;=\; \|z_u\|\,\|z_v\| \cos\theta_{uv},$$

and the two lengths are fixed once the drugs are fixed, so the only thing that varies from pair to pair is $\cos\theta$. For the model to separate interacting from non-interacting pairs, the angles have to vary.

Table 8 reports the geometry of the four feature spaces: the mean length of a drug vector, and the mean cosine between two of them.

| Features | Mean norm | Mean $\cos\theta$ | Mean angle |
|---|---|---|---|
| Random vectors | 27.7 | 0.000 | 90° |
| ChemBERTa | 26.7 | 0.670 | 48° |
| GPT (ada-002) | 1.0 | 0.756 | 41° |
| PubMedBERT | 15.1 | 0.921 | 23° |

Table 8: Geometry of the feature spaces, over 20,000 random pairs of distinct drugs. The GPT vectors are returned already normalised.

The random vectors point in all directions, as one expects from Gaussian noise in 768 dimensions. The embeddings do not: in PubMedBERT space any two drugs are within about 23° of each other. The vectors have 768 dimensions but sit inside a narrow cone, so there is very little angular variation left for the decoder to use. This property is called anisotropy and is known to affect the spaces produced by BERT-like models [2].

The order of the four spaces, $0.000 < 0.670 < 0.756 < 0.921$, is exactly the order of the four rows in Table 7, reversed, in both training regimes. The less the vectors vary in direction, the worse the model does. The GPT row is a prediction rather than an observation: we measured its cosine before running it, found 0.756, and expected it to land between ChemBERTa and PubMedBERT. It did.

One further detail points the same way. For PubMedBERT the length of a drug's vector is negatively correlated with its degree (Spearman $\rho = -0.22$): the best connected drugs get the shortest vectors, so the features push exactly those pairs towards lower scores, against what the graph structure says. The random vectors show no such correlation ($\rho = 0.01$).

**Removing the anisotropy**

So far we only have a correlation: the embeddings whose vectors point in similar directions do worse. To test it, we removed the concentration directly.

The transformation is called whitening. The idea is the one behind a z-score, where a value is centred by subtracting the mean and rescaled by dividing by the standard deviation, so that the result says how far from typical it is. Whitening does the same to vectors. First we subtract the mean vector: that mean is the common direction every drug was pointing in, and removing it does most of the work. Then we find the directions in which the drugs actually vary and divide each by its own standard deviation, so that no direction counts more than another. Finally we rotate the result back to the original orientation, which keeps each vector close to where it started. This last step is what makes the transformation ZCA rather than PCA whitening [7]; for our model it changes little, since a fixed rotation of the input can be absorbed into the weights of the first layer, but it keeps the whitened vectors comparable with the original ones. We then rescale everything so that the mean norm matches the random vectors, leaving direction as the only difference.

Written out, with $X$ the matrix of embeddings, $X_c = X - \bar{x}$ the centred version and $\Sigma = V\Lambda V^\top$ the eigendecomposition of its covariance:

$$X_w \;=\; X_c\, V\,(\Lambda + \epsilon I)^{-1/2}\, V^\top, \qquad \epsilon = 10^{-5}.$$

The covariance of $X_w$ is the identity, so no direction is favoured.

Two points are worth stressing. The transformation adds no information: it is one fixed linear map, the same for every drug, computed from the embeddings alone, without labels, without the graph and without the test data. And it is not new: the same correction has been applied to sentence embeddings for the same reason [12].

It works. The mean cosine falls from 0.921 to 0.020 for PubMedBERT and from 0.670 to 0.027 for ChemBERTa.

The effect on the results is large. On the degree-matched set with random negatives, PubMedBERT goes from 0.642 to 0.871 and ChemBERTa from 0.690 to 0.860. Both whitened embeddings now beat the random vectors, and whitened PubMedBERT is the best single input in all six columns. The features do carry information about drug interactions; the dot product could not reach it.

Two things did not turn out as expected. We thought the gain would follow the amount of anisotropy removed, since PubMedBERT starts far more concentrated. It does not: PubMedBERT gains 0.229 and ChemBERTa 0.170, but they finish within 0.011 of each other, less than one standard deviation. The anisotropy behaves like a threshold rather than a dial.

We also checked that whitening had not collapsed the two embeddings onto the same representation, which would make their equal performance an artefact. For each drug we took its ten nearest neighbours in each whitened space: the two lists share 0.023 of their members on average. The spaces disagree almost completely about which drugs resemble which, and still give the same AUROC. On this dataset, chemical structure and clinical text are two different and equally usable views of the drugs.

### Combining the two embeddings

Since the two whitened spaces encode different things, we tested whether they are complementary, by concatenating them into a 1,536-dimensional input. Doubling the input also doubles the parameters of the first convolution layer, so we ran a 1,536-dimensional random Gaussian input as a control: same capacity, no information.

Table 9 reports the concatenation against the better of the two embeddings alone and against a random input of the same dimension.

The control matters. Going from 768 to 1,536 random dimensions is worth +0.028 on its own, with no information added. Against that baseline the concatenation gains +0.027, close to the +0.040 that whitened PubMedBERT gains over the 768-dimensional random vectors. The two embeddings together are therefore not clearly better than the better of the two alone, once capacity is accounted for. This also means the rows of Table 7 are not a single ranking:

| Features | dim | Degree-matched | CRESCENDDI |
|---|---|---|---|
| Random vectors | 768 | $0.831 \pm 0.003$ | $0.625 \pm 0.009$ |
| PubMedBERT whitened | 768 | $0.871 \pm 0.014$ | $0.645 \pm 0.014$ |
| Random vectors | 1,536 | $0.859 \pm 0.011$ | $0.627 \pm 0.008$ |
| Both whitened, concatenated | 1,536 | $0.886 \pm 0.010$ | $0.661 \pm 0.010$ |

Table 9: Concatenating the two whitened embeddings, against a random input of the same dimension. Trained on random negatives, 5 seeds.

each should be compared with a random input of its own size, which is why the GPT row cannot be read directly against the others.

**What this means**

The project began by asking whether biomedical embeddings improve on general-purpose ones. The answer is more specific than the question. Used as they come, with a dot-product decoder, they are worse than random vectors on every evaluation set that is not the benchmark, and the ordering across three different embeddings follows their geometry rather than their content. A linear transformation that adds no information removes the problem, after which both embeddings beat the random vectors and the difference between text and structure disappears.

One consequence reaches beyond our own setting. A study that reports an improvement from semantic features, or the absence of one, may be reporting a property of the embedding geometry rather than of the information the embedding contains. The two are easy to confuse when only one baseline is used.

## 4.6 Without the graph

The degree-only model of Section 4 uses the graph and knows nothing about the drugs. Here we do the opposite: the classifiers of Section 3.2 see the node features and no structure at all.

On the official split these models score well without knowing anything. A logistic regression given only the identities of the two drugs reaches 0.979 on the CRESCENDDI test set, and a random forest on random vectors reaches 0.822, while both are at chance on the two other sets. They recognise the 430 drugs the verified pairs are built from, which is enough because 807 of the 811 test pairs use them. Everything below therefore uses the drug-disjoint split of Section 3.3, where the same drug-identity model returns exactly 0.500, averaged over three splits.

Table 10 reports the classifiers on the drug-disjoint split, with two reference rows: the degree-only model, which uses the graph and no features, and random vectors, which use neither.

| Features | Model | Random | Degree-matched | CRESCENDDI |
|---|---|---|---|---|
| Degree only (graph, no features) | LR | $0.165 \pm 0.008$ | $0.693 \pm 0.014$ | $0.695 \pm 0.012$ |
| Random vectors | LR | $0.500 \pm 0.001$ | $0.491 \pm 0.005$ | $0.506 \pm 0.018$ |
| Random vectors | RF | $0.498 \pm 0.004$ | $0.506 \pm 0.023$ | $0.500 \pm 0.012$ |
| PubMedBERT | LR | $0.333 \pm 0.003$ | $0.423 \pm 0.007$ | $0.763 \pm 0.020$ |
| PubMedBERT | RF | $0.279 \pm 0.006$ | $0.364 \pm 0.027$ | $0.826 \pm 0.008$ |
| PubMedBERT whitened | LR | $0.501 \pm 0.003$ | $0.497 \pm 0.009$ | $0.509 \pm 0.010$ |
| PubMedBERT whitened | RF | $0.396 \pm 0.007$ | $0.410 \pm 0.029$ | $0.531 \pm 0.023$ |

Table 10: Classifiers on the node features, with no message passing, trained on the CRESCENDDI negatives, on the drug-disjoint split. Means and standard deviations over 3 splits. The degree-only row is a reference: it uses the graph and no features, and is unaffected by the split, since the degree is a property of the graph.

The degree-only row shows that the split does not remove the degree effect, and was not meant to: trained on verified negatives, the rule "two well connected drugs are safe" still gives 0.695 on CRESCENDDI, and 0.165 on the random negatives, where it points the wrong way.

Against that, the whitened embeddings, which are the best input for the GCN, give a logistic regression at most 0.014 over random vectors, in any column of either training regime. Raw PubMedBERT is high on CRESCENDDI alone (0.763 for the logistic regression, 0.826 for the random forest) and below chance on the two other sets; that value is taken apart below. Trained on random negatives instead, the same models follow the degree, with raw PubMedBERT reaching $0.771 \pm 0.006$ on the random negatives and $0.352 \pm 0.013$ on CRESCENDDI.

**Where a value of 0.900 comes from**

Before taking the largest value apart, one more property of the data is needed. The CRESCENDDI pairs are built from 430 drugs listed in clinical compendia, so they are marketed and widely prescribed, while the positive edges come from all of DrugBank, where many drugs are experimental. A drug description shows which of the two a drug is. A model can therefore guess the label from the drugs alone, without judging the pair, and this remains available after the drug-disjoint split, since it is a property of the kind of drug rather than of the drug itself.

On the official split, a logistic regression on raw PubMedBERT trained on CRESCENDDI reaches 0.901 on the CRESCENDDI test pairs, the value Li et al. [11] report for a graph neural network on this dataset, obtained here without the graph. It rests on two effects. Holding the test drugs out of training costs 0.138. Restricting the positives to pairs of CRESCENDDI drugs, so that both classes come from the same population with similar degrees, costs a further 0.198 and leaves $0.565 \pm 0.022$, against $0.551 \pm 0.009$ for a model that sees only the degrees

of the same pairs. The random forest keeps more, $0.639 \pm 0.028$, on each of the three splits; we cannot say whether that is a property of the pair or a further property of the single drugs, and report it as an open residual.

Used without the graph, then, the embeddings carry information about single drugs: which drug it is, and which population it belongs to. Neither says whether these two drugs interact.

# 5 Discussion

## 5.1 What the benchmark measures

A logistic regression with two features reaches 0.880 on the standard evaluation, against 0.905 for a GCN with 768-dimensional biomedical embeddings. Since an AUROC of 0.5 is what a coin flip gives, that is 94% of the GCN's value above chance, obtained from the degrees of the two drugs and nothing else. The GCN also produces most of its benchmark value after a single gradient step. Both results say that the number reported under the standard protocol is largely a measurement of how connected the drugs are, not of whether they interact.

This is not a property of our model. It follows from three things. The negatives are sampled uniformly, so they are much less connected than the positives. Message passing makes the representation of a node grow with its number of neighbours, and the dot product of two long vectors is large. And the features, as they come, vary so little in direction that the lengths are the main thing the decoder can read. The third ingredient is the one that can be changed: after whitening, the GCN reaches 0.871 on the degree-matched set, so the behaviour is not inevitable.

## 5.2 Verified negatives reverse the shortcut

Training on the CRESCENDDI pairs does not remove the degree effect. The verified negatives are more connected than the positive edges, so the rule becomes "two well connected drugs are safe". On its own that rule gives 0.684 on CRESCENDDI and 0.695 on the degree-matched set, against 0.147 on the random negatives, where it points the wrong way.

What the verified negatives do give is stability. The GCN trained on them varies by 0.080 between its three evaluation values, against 0.420 when it is trained on random negatives and 0.548 for a degree rule trained on the same pairs. The configuration with the best benchmark value is the one that varies most, so choosing a configuration on its benchmark value selects against robustness on this dataset.

## 5.3 Geometry, not content

The feature comparison adds a result we did not expect. Used as they come, all three embeddings do worse than 768 random numbers per drug on both control sets, and their order follows their geometry rather than their content.

A whitening transformation that adds no information reverses this, after which whitened PubMedBERT is the best input in every column.

This matters beyond our own setting. A study that reports an improvement from semantic features, or the absence of one, may be reporting a property of the embedding geometry rather than of the information the embedding contains. The two are easy to confuse when the only baseline is another embedding. A random input of the same dimension is cheap and settles the question.

## 5.4 Drug-level and pair-level signals

Our controls found three ways of scoring well without judging a pair. The degree says how connected a drug is. Drug identity says which drug it is: a model given only the names of the two drugs reaches 0.979 on the CRESCENDDI test set, and a random forest on random vectors reaches 0.822, both by recognising the 430 drugs the verified pairs are built from. The population says what kind of drug it is: the CRESCENDDI pairs come from marketed clinical drugs while the positive edges come from all of DrugBank, and a drug description shows which is which.

All three describe single drugs, while the task asks about a pair. An evaluation in which the same well studied drugs appear everywhere rewards drug-level signals, and the three we found are enough to produce most of the values we measured. Taking the largest of them apart shows this directly: a logistic regression reaching 0.901 on CRESCENDDI falls to 0.763 when the test drugs are held out of training, and to 0.565 when the positives are restricted to CRESCENDDI drugs, against 0.551 for a model that sees only the degrees.

Against that background, one result has no drug-level explanation. With whitened features the GCN reaches 0.889 on the degree-matched set, where a degree rule gives at most 0.695, where the drugs are not those the model was trained on, and where the same features, evaluated without the graph on drugs never seen in training, stay within 0.014 of random vectors. Whatever the GCN uses there is a property of the pair, and it comes from the graph.

## 5.5 Relation to existing work

Yang et al. [14] already argued that the way negatives are handled makes reported results hard to compare. More recently, Li et al. [10] re-evaluated several methods under a unified protocol and found that their results on `ogbl-ddi` differ substantially from the published ones, in both directions. They further show that on this dataset validation and test performance are only weakly related under the standard evaluation, to the point where validation performance is a poor guide for tuning [10, Appendix E]. This is related to the saturation we observe, although in their case both sets use random negatives, while ours compares a validation set of random negatives with control sets built differently. Their answer is to sample harder negatives heuristically. Ours is different in one respect: the CRESCENDDI negatives are not harder by construction, they are verified, so a failure on them is a failure on pairs we know to be non-interacting.

Verified negatives bring a difficulty of their own, worth stating for anyone who wants to use them. They are curated, so they come from a particular set of drugs, and those drugs appear in both the training and the test pairs unless one takes care. Whenever the positives and the negatives are drawn from different sources, a model can learn the source instead of the label. Two controls are enough to check it: hold the test drugs out of training, and draw the positives from the same drugs as the negatives.

Our results also bear on a companion project in the same group. Giobbi [3] predicts drug–drug interactions on a graph built from ChCh-Miner and CRESCENDDI, with GPT `text-embedding-ada-002` embeddings of the drug descriptions as node features, and reports that semantic enrichment improves on a non-informative feature baseline. We do not contradict that finding, and the comparison against uninformative features is exactly the control we argue for. Two differences in the setting matter, though. ChCh-Miner has 1,514 drugs and 48,514 edges, a density of about 4% against 14.7% here, so the gap in degree between positive edges and uniformly sampled negatives is much smaller, and the same CRESCENDDI mapping produces an overlap of only 4.6% instead of 55.1%. The mechanism we describe needs a dense graph with a skewed degree distribution, which `ogbl-ddi` has and ChCh-Miner does not to the same extent.

What does transfer is the set of controls. On `ogbl-ddi`, raw GPT embeddings are no better than random vectors on the benchmark (Table 7) and worse on both control sets, so a gain measured only against uninformative features on the standard evaluation does not by itself show that the semantic content is being used. And the CRESCENDDI drugs form a larger share of a 1,514-node graph than of ours, so a drug-disjoint check (Section 3.3) is at least as necessary there.

# 6 Limitations

## 6.1 The method cannot handle a drug it has never seen

Our setting is transductive: all 4,267 drugs are in the graph during training, and the model computes a representation for each of them, including those that only appear in test edges. This is the standard protocol [6], but the restriction is not only one of evaluation.

A GCN builds the representation of a drug by aggregating over its neighbours. A drug that has just been approved has no edges, so there is nothing to aggregate, and the encoder can only use the drug's own features. Section 4.6 measures what those features are worth alone: on drugs never seen during training, and without the graph, a logistic regression on the whitened embeddings stays within 0.014 of one using random vectors. The prediction for an isolated drug would be close to chance.

This matters for the application. The drugs where an interaction prediction would be most useful are exactly the recent ones, where the clinical record is thin and an unexpected interaction is most likely to cause harm. A method

that needs a drug to be already well connected is least usable where it would be most valuable. The same is true of the degree effect: a model that relies on connectivity gives low scores to poorly studied drugs, which is the right answer about the recorded data and the wrong answer about the clinical question.

Predicting for a new drug would need two changes. The architecture would have to aggregate the neighbours' features rather than their identities, as GraphSAGE does [5], and the training would have to hide some drugs from the graph, so that the model is made to use the features. Even then, the features would have to exist before the drug is studied, which points to the chemical structure rather than the text description; after whitening, ChemBERTa is as good as PubMedBERT (0.860 against 0.871).

## 6.2 What our results do and do not show

What we show is a property of the evaluation, not a ranking of methods. On this dataset, a large part of what the standard protocol measures is available from the degree alone, is present before training, and does not survive a change in how the negatives are built. That says something about the benchmark. It does not say that graph neural networks are ineffective for interaction prediction, and it does not tell anyone which method to use.

Two consequences. First, our conclusions concern `ogbl-ddi`, and more generally evaluations built on uniformly sampled negatives on dense graphs with skewed degrees. Whether they transfer to sparser graphs is open, and the BioSnap comparison above suggests they may not. Second, we tested one architecture with one decoder. A decoder with its own parameters, or an architecture that normalises the node representations, might be less exposed to the degree effect than the dot product we use. We did not measure this.

## 6.3 Remaining risks

Three points from Section 3 are not settled by the controls we put in place.

The features come from a language model pretrained on the same literature the interactions were curated from, so part of what we ask the model to predict may already be in its inputs. We report this as a risk rather than a measured quantity, because measuring it would need a language model trained with the DDI literature removed.

The CRESCENDDI evaluation set has 811 pairs, which is small. The standard deviation across seeds there is 0.019, about seven times the one on the random negatives, and the model sits near 0.5, where AUROC is least stable. The differences we report between training configurations are much larger than that, but the individual values should be read with this width in mind.

The degree-matched control matches the degrees of the drugs, but not the drugs themselves. The CRESCENDDI pairs come from the 430 drugs of the clinical compendia, while the degree-matched pairs are drawn from the whole graph at the same degrees. Section 4.6 shows that this difference is usable: on the drug-disjoint split, a logistic regression on raw PubMedBERT trained on

random negatives, which never saw a CRESCENDDI label, separates the two sets by 0.262, from 0.614 on the degree-matched set to 0.352 on CRESCENDDI. Where we compare the two sets, the difference we attribute to the clinical verification therefore also contains an effect of which drugs the sets are made of. A control matched on the drug population, in which the positives are restricted to pairs of CRESCENDDI drugs, separates the two; we use it in Section 4.6 for the classifiers but not for the GCN, which would require new runs.

# 7 Future work

Four things follow directly from the results above.

The drug-disjoint split should be applied to the GCN. For a classifier the split is a filter on the pairs, but a GCN also passes messages along the edges, so the edges of the held-out drugs would have to leave the graph as well. That is an inductive setting rather than the transductive one used here, and it would show how much of what the GCN achieves on the CRESCENDDI pairs survives when the drugs are new.

The population-matched evaluation should be extended to the GCN. Both classes are then made of the same drugs with similar degrees, and what remains between them is the clinical verification. This is the cleanest comparison we have for what the verification contributes, and it applies to any model. It would also settle the residual the random forest keeps over the degree model in Section 4.6. A first step would be to re-pair the drugs at random while keeping each drug's vector, which leaves every drug-level property intact and destroys every pair-level one.

The degree is not the only structural quantity that separates positive edges from sampled non-edges. A control matched on the number of common neighbours, in addition to the degree, would tighten the argument. This matters more on `ogbl-ddi` than elsewhere: because of its density, most randomly sampled non-edges here do share common neighbours, unlike in the sparser benchmark datasets [10].

Finally, the imbalance between the two classes could be handled better. In the CRESCENDDI regime we subsample the positive edges to a 10:1 ratio, which discards most of them, and the subsample changes with the seed. Training in mini-batches would let the model see all the positive edges while reusing the verified negatives across batches, at the cost of running the encoder on the full graph at every step. It creates no new negatives, though: there are 2,834 of them, built from 430 drugs, so more exposure to the same pairs makes the drug-disjoint check more important rather than less.

# 8 Conclusion

We set out to ask whether biomedical embeddings and clinically verified negatives improve drug–drug interaction prediction on `ogbl-ddi`. Both answers

depend on a third thing we had to measure first: what the benchmark rewards.

On the standard evaluation, most of what is measured is the degree. Two numbers per pair give 0.880 against 0.905 for a GCN with 768-dimensional features, and the same shortcut is present after one gradient step, before the model has learned anything. It comes from the data, where the sampled negatives involve far less studied drugs than the positives, and from message passing, which makes the representation of a well connected drug longer.

The embeddings do help, but not as they come and not on their own. Raw PubMedBERT and ChemBERTa are worse than random vectors on both of our control sets, because their vectors are nearly parallel and our decoder reads only angles. Whitening, a fixed linear map that adds no information, takes PubMedBERT from 0.642 to 0.871 on the degree-matched set and makes it the best input in every column. Without the graph, however, the same features are worth no more than random numbers. The information they carry is usable only in combination with the structure.

The verified negatives do not remove the degree effect, they reverse it, since the drugs they involve are more connected than the positives. Their real benefit is stability: the model trained on them varies by 0.080 between evaluation sets, against 0.420 for the model trained on random negatives. A configuration chosen on its benchmark value is therefore not the most robust one.

Verified negatives also bring a risk of their own. They come from a small set of well studied drugs, which appear in both the training and the test pairs, and a model can score them by recognising the drugs. A model given only the two drug identities reaches 0.979, and a random forest on random vectors reaches 0.822; both fall to 0.500 when the test drugs are held out of training.

One result survives every control we built. With whitened features, the GCN reaches 0.889 on the degree-matched set, where a degree rule gives at most 0.695, where the drugs are not those seen in training, and where the same features without the graph are no better than random vectors. Whatever the model uses there is a property of the pair, and it comes from the graph.

What we would do next follows from the same logic: apply the drug-disjoint split and the population-matched evaluation to the GCN, and match the controls on the number of common neighbours as well as on the degree (Section 7).

None of this means that graph neural networks fail at interaction prediction. Our own model does better than any shortcut we could find, once its features are whitened. What it means is that a single number on this benchmark cannot tell you whether a model has learned anything, and that the controls which can are easy to run.

# References

[1] Seyone Chithrananda, Gabriel Grand, and Bharath Ramsundar. Chemberta: Large-scale self-supervised pretraining for molecular property prediction. *arXiv preprint arXiv:2010.09885*, 2020.

[2] Kawin Ethayarajh. How contextual are contextualized word representations? comparing the geometry of BERT, ELMo, and GPT-2 embeddings. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 55–65, Hong Kong, China, 2019. Association for Computational Linguistics. doi: 10.18653/v1/D19-1006.

[3] Luca Giobbi. Detecting drug interactions with machine learning. DAS in data science report, ETH Zurich, May 2026.

[4] Yu Gu, Robert Tinn, Hao Cheng, Michael Lucas, Naoto Usuyama, Xiaodong Liu, Tristan Naumann, Jianfeng Gao, and Hoifung Poon. Domain-specific language model pretraining for biomedical natural language processing. *ACM Transactions on Computing for Healthcare*, 3(1):1–23, 2021.

[5] William L. Hamilton, Rex Ying, and Jure Leskovec. Inductive representation learning on large graphs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[6] Weihua Hu, Matthias Fey, Marinka Zitnik, Yuxiao Dong, Hongyu Ren, Bowen Liu, Michele Catasta, and Jure Leskovec. Open graph benchmark: Datasets for machine learning on graphs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[7] Agnan Kessy, Alex Lewin, and Korbinian Strimmer. Optimal whitening and decorrelation. *The American Statistician*, 72(4):309–314, 2018. doi: 10.1080/00031305.2016.1277159.

[8] Thomas N. Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. In *International Conference on Learning Representations (ICLR)*, 2017.

[9] Elpida Kontsioti, Simon Maskell, Bhaskar Dutta, and Munir Pirmohamed. A reference set of clinically relevant adverse drug-drug interactions. *Scientific Data*, 9(1):72, 2022. doi: 10.1038/s41597-022-01159-y.

[10] Juanhui Li, Harry Shomer, Haitao Mao, Shuaiqiang Zeng, Yao Ma, Neil Shah, Jiliang Tang, and Dawei Yin. Evaluating graph neural networks for link prediction: Current pitfalls and new benchmarking. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[11] Xin Li et al. Llm-ddi: Leveraging large language models for drug-drug interaction prediction on biomedical knowledge graph. *IEEE Journal of Biomedical and Health Informatics*, 30(1):778–789, 2026.

[12] Jianlin Su, Jiarun Cao, Weijie Liu, and Yangyiwen Ou. Whitening sentence representations for better semantics and faster retrieval. *arXiv preprint arXiv:2103.15316*, 2021.

[13] David S. Wishart, Yannick D. Feunang, An C. Guo, et al. Drugbank 5.0: a major update to the drugbank database for 2018. *Nucleic Acids Research*, 46(D1):D1074–D1082, 2018.

[14] Yang Yang, Ryan N. Lichtenwalter, and Nitesh V. Chawla. Evaluating link prediction methods. *Knowledge and Information Systems*, 45(3):751–782, 2015. doi: 10.1007/s10115-014-0789-0.